

Comparing and Interpreting Models

Quantitative Reasoning II · Lesson 8

Below are five different simple linear regression models, all run on the same dataset: 2,925 home sales in Ames, Iowa. The dependent variable in all models is the sale price of the home in dollars. Each column reports the results of a simple linear regression with one independent variable.

	Living area	Lot area	Age at sale	Garage	Quality
Description of X	above-grade sq ft	lot sq ft	years since built	car capacity	finish, 1–10
Range of X	334–3,820	1,300–215,245	0–136	0–5	1–10
n	2,925	2,925	2,925	2,924	2,925
Intercept	\$6,773	\$152,869	\$233,929	\$61,329	−\$94,088
Slope	\$116.23	\$2.73	−\$1,467	\$67,473	\$45,087
R^2	0.518	0.073	0.320	0.426	0.648

1. For each model write one sentence interpreting the slope of the model (including units!)
2. Which of the five models explains the most variation in price? Which explains the least?